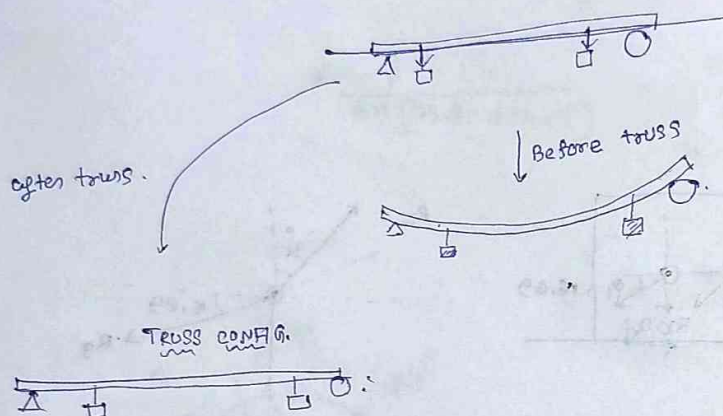
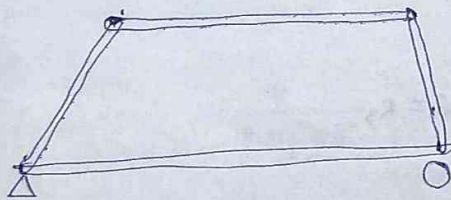


Truss

- it is a structure made of several members and it is used to support the load
- Purpose: - support the load and convert the transverse load to axial load
- remove the deflections present in the load



BASIC-SHAPE OF TRUSS ??



→ rigid and non-collapsible
in nature.

→ triangular in shape

→ there must be a diagonal
member present.

TRUSS

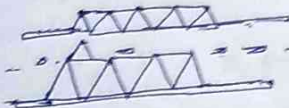
PLAN TRUSS

→ all members are present in the same plane



SPACE TRUSS

→ members are present in diff-2 planes



→ bridge truss

NH 15

eg mobile towers etc

ANALYSIS OF TRUSS

→ analyse in truss members are no always solution

Assumptions of truss :- ① The members will be connected at the ends

② All members will be connected through frictionless pin joints.

③ The loads will be applied only at the joints

④ The members will either experience tensile or compressible in nature. This thing is called two forces member.

⑤ Weight of members are negligible

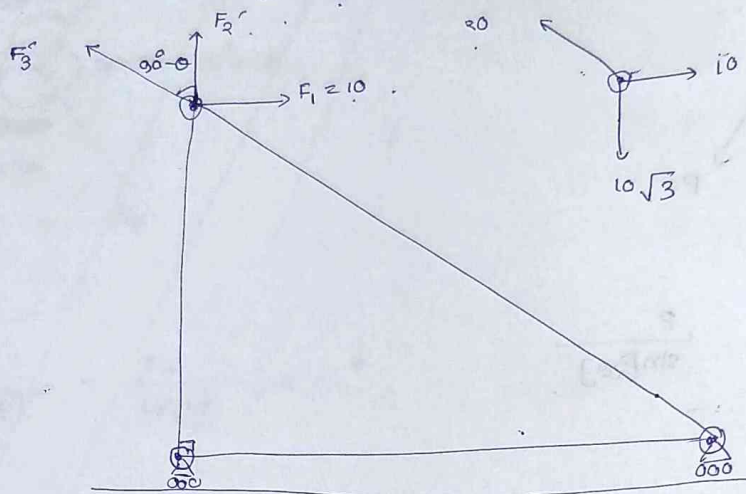


Method of Joint

→ consider all forces as either
tensile or compressible but

not both

Note:-
each joint
has two eqⁿ eq^s



$$F_2 + F_3 \sin \theta = 0$$

$$F_1 - F_3 \cos \theta = 0$$

↓

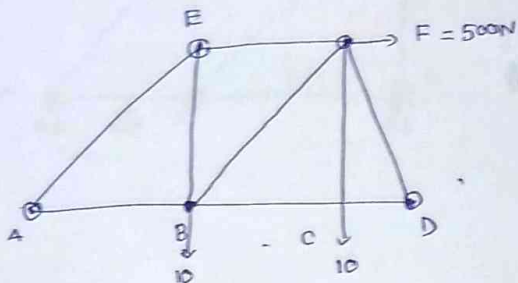
$$+20 = F_3$$

$$F_2 + (+20) \frac{\sqrt{3}}{2} = 0$$

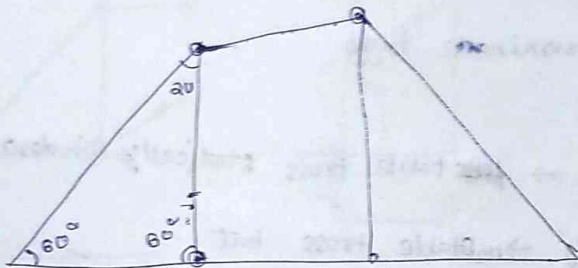
$$F_2 = -10\sqrt{3}$$

→ identifying 0 force member

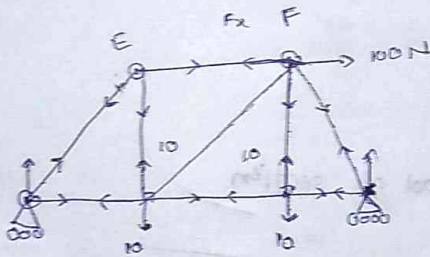
perfect Truss



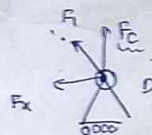
length of each member = 2m
 $AB = BE = 2$
 $BC = CF = 2$
 $CD = 2$



- ① First calculate the reaction force. (as it is shared external force)
- ② check is there any 0 force members
- ③ Apply the "method of joint" concept



Apply $\sum F = 0$ for whole system as system is in rest



$$\frac{500 \times 2}{3} = R_D$$

$$R_D = \frac{530\sqrt{3}}{3}$$

$$F_x = \frac{530}{6}$$

Answer



Perfect Truss derivation

Q.

m = number of members

J = number of joints

* Number of unknowns = $(m+3)$

No of eq equation = $2J$

$$\boxed{m+3 = 2J}$$

→ stable truss

→ statically determinate truss

If $m+3 > 2J$ → ~~stable~~ truss. statically indeterminate

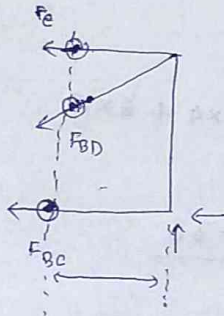
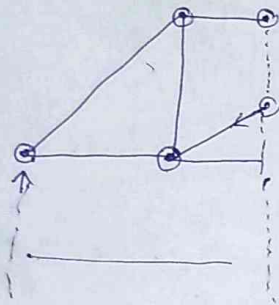
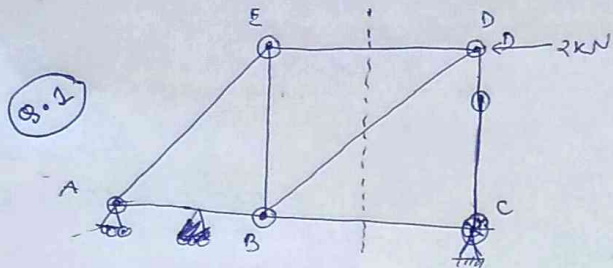
If $m+3 < 2J$ → unstable truss but

statically indeterminate

Method of section

Q2 Method of Section (examples + explain)

~~Legend Technology~~

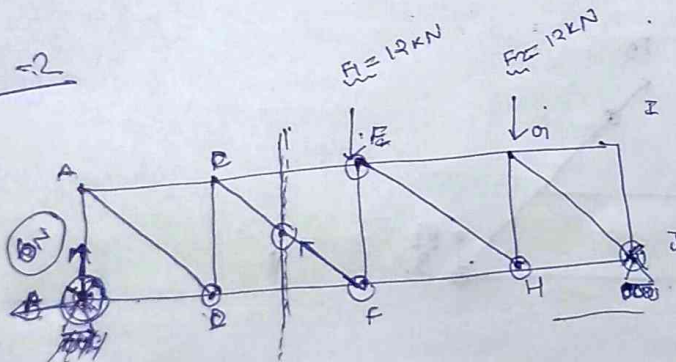


FBD

$$\begin{array}{r} 6 \\ 2 \\ \hline 8 \\ 46 \\ \hline \Sigma F_x \end{array}$$

$$\frac{F_{BD}}{\sqrt{2}} = 10 \Rightarrow F_{BD} = 10\sqrt{2}$$

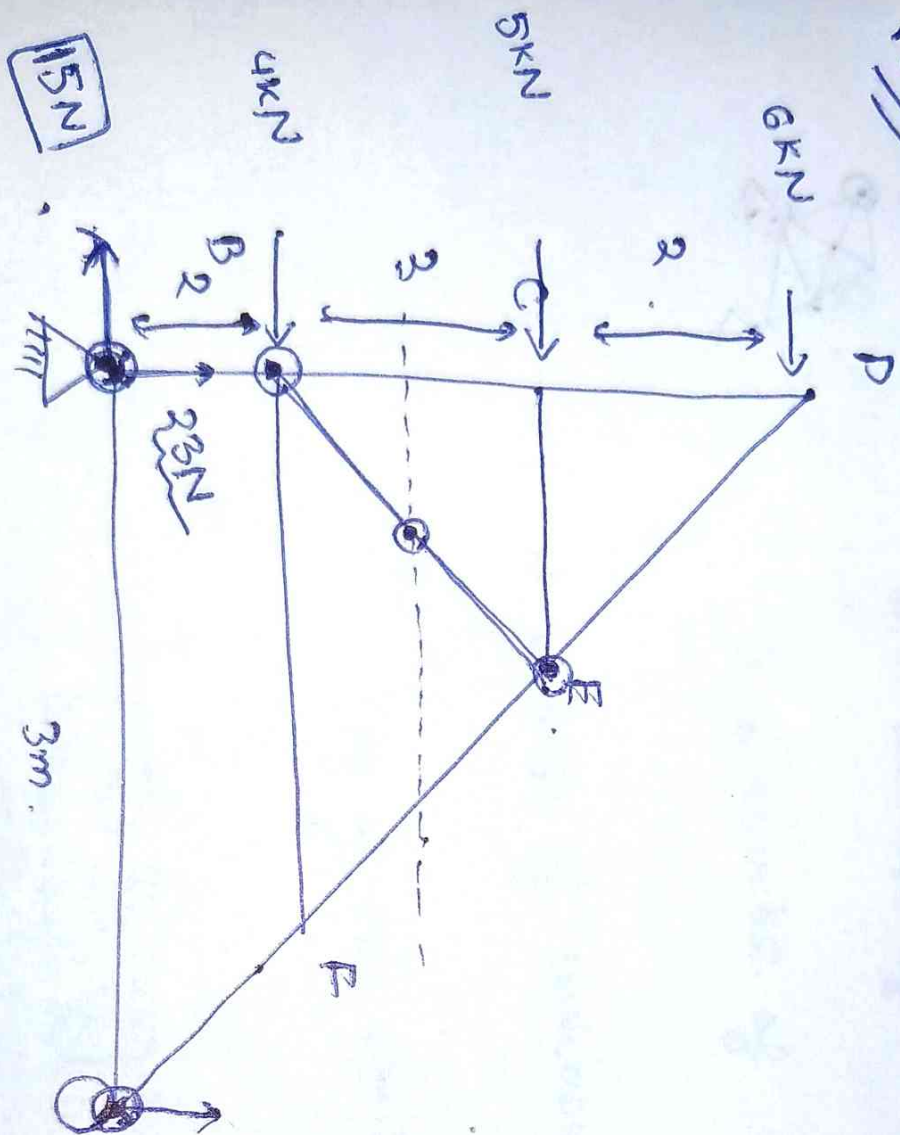
Problem 2



$$F_{BC} = 6\sqrt{2}$$

$$\begin{aligned} 12 \times 2 + 12 \times 1 &= F \times 4 \\ \frac{3}{12} \times \frac{3}{4} &= F \\ 9 &= F \end{aligned}$$

Problem 3



calculation for at BE